

Flat Boundaries Preserve Information

Sharp recovery of uniform convolution factors
without Gaussian smoothing

Research report prepared for Vladimir Reshetnikov

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Abstract

A preceding ProveIt report established total-variation stability for finite uniform-factor deconvolution but left statistical lower bounds unresolved. A companion report obtained sharp rates under a strictly positive Gaussian variance floor and explicitly asked whether that floor could be removed for the Fabius–Rvachev density. We supply the missing analytic argument. For every infinite summable convolution of positive-width centered uniforms, and every pair of perturbation laws having all absolute moments finite, a first unmatched moment of order r gives an exact squared Hellinger asymptotic of order h^{2r} , with constant $\Delta_r^2 \int (f^{(r)})^2 / f / (4(r!)^2)$. Finite uniform prefixes, conditional-score contraction, and convergence of higher-order information provide a proof that controls the moving support rather than ignoring it.

Consequently the sharp scale-recovery rate is $n^{-1/(4m)}$ with known Gaussian variance, including variance zero. With unknown variance in $[0, V]$, the scale and variance rates are $n^{-1/(4m+4)}$ and $n^{-1/(2m+2)}$; matching hard pairs approach the zero-noise boundary and one member has exactly zero Gaussian variance. We also prove a finite-prefix boundary trichotomy, including an explicit logarithmic critical case, and exhibit identifiable perturbations whose Hellinger separation is smaller than every power of their amplitude. The article includes full conventional proofs, reproducible checks, and a research agenda. No proof-assistant verification or worldwide publication-priority claim is made.

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1 Research target, provenance, and contribution boundary

The question is not whether moment cancellation occurs. It does. The question is whether cancellation survives division by a density that vanishes to all orders at the endpoints of its support. This is the step needed to turn an algebraic inverse modulus into a statistically sharp lower bound.

1.1 The specific gap being continued

At the pinned repository snapshot, the report *Sharp Stability Strata for Finite Fabius–Rvachev Deconvolution* proves local total-variation inverse exponents, exact density expansions, and a sampling upper bound [6]. Its source, lines 1190–1234, explicitly declines to call that upper bound statistically minimax and identifies Hellinger or relative entropy control as the missing ingredient. The global scale upper bound has exponent $1/(4m)$ when the capacity is m .

A separate, earlier-today companion manuscript, *Gaussian Confounding and Sharp Recovery of Uniform Convolution Factors*, proves corresponding statistical rates with a strictly positive Gaussian variance floor [7]. Its Research Question 1, page 17, asks for the unsmoothed up-law lower bound and specifically for control of $\int (g^{(k)})^2/g$ and the changing support. That manuscript was read from the user’s Library, not asserted to be a file in the pinned repository.

The present article resolves this question for the up density and, more generally, for every infinite summable positive-width uniform convolution. It also proves that the unknown-variance lower-bound obstruction persists arbitrarily close to zero Gaussian variance. This last assertion is stronger than a global lower bound obtained merely by restricting to a positive-noise submodel. It is not a claim that every mixed collision stratum is classified.

1.2 What is classical and what is developed here

The Fabius–Rvachev infinite-product construction is classical; see Arias de Reyna [1]. Higher-order Fisher information is also classical. Bobkov’s treatment [2] discusses its origin, convolution monotonicity, and sufficient conditions for finiteness. In particular, all-order information finiteness for an infinite uniform convolution is not presented here as a newly invented functional or an isolated breakthrough. We give the elementary spline proof needed by our argument.

Moment matching and singular moment inversion have substantial statistical precedents, including the mixture-estimation work of Heinrich–Kahn and Wu–Yang [3, 4]. Our parameters describe a *convolution of independent factors*, not a mixture with free weights. Those papers provide context, not substitute proofs for this model. The shifted-power-sum inequality in section 8 is credited to the companion manuscript and reproduced to make the statistical conclusions self-contained.

The main development here is the exact all-order Hellinger theorem at a flat moving boundary, its finite-prefix boundary trichotomy, and the resulting zero-noise statistical lower bounds. The proofs are mathematical arguments in this article, not kernel-checked results. The literature comparison is focused; it establishes a precise continuation of the identified reports, not exhaustive worldwide novelty.

2 Definitions and main results

Let $(U_j)_{j \geq 1}$ be independent uniforms on $[-1, 1]$. Fix

$$b_j > 0, \quad S = \sum_{j \geq 1} b_j < \infty, \quad B = \sum_{j \geq 1} b_j U_j. \quad (2.1)$$

All infinite sums here converge absolutely for every outcome. Write f for the density of B , and f_N for that of $B_N = \sum_{j=1}^N b_j U_j$. We will prove the smoothness and positivity facts used below. The

Fabius–Rvachev specialization is

$$b_j = 2^{-j}, \quad S = 1, \quad f = \text{up}. \quad (2.2)$$

With the characteristic-function convention $\mathbb{E}e^{itB}$, it has product $\prod_{j \geq 1} \text{sinc}(2^{-j}t)$.

Our Hellinger normalization, throughout, is

$$H^2(p, q) = \int_{\mathbb{R}} (\sqrt{p} - \sqrt{q})^2 dx, \quad \rho(p, q) = \int_{\mathbb{R}} \sqrt{pq} dx = 1 - \frac{1}{2}H^2(p, q). \quad (2.3)$$

Thus $0 \leq H^2 \leq 2$ and $\text{TV}(p, q) = \frac{1}{2} \int |p - q| \leq H(p, q)$. For a probability law ν , let ν_h denote the law of hZ when $Z \sim \nu$, with $h > 0$. A law is *all-moment* if $\int |z|^k d\nu(z) < \infty$ for every integer $k \geq 1$. An exponential moment is not required.

Theorem 2.1 (Exact Hellinger moment asymptotic). *Let f be defined by (2.1). For every integer $r \geq 1$,*

$$0 < J_r(f) := \int_{f>0} \frac{(f^{(r)}(x))^2}{f(x)} dx < \infty. \quad (2.4)$$

Suppose ν, η are all-moment laws whose moments of orders $0, \dots, r-1$ agree, and set $\Delta_r = \int z^r d\nu - \int z^r d\eta$. Then

$$H^2(f * \nu_h, f * \eta_h) = \frac{\Delta_r^2}{4(r!)^2} J_r(f) h^{2r} + o(h^{2r}). \quad (2.5)$$

More precisely, in $L^2(\mathbb{R})$,

$$\frac{\sqrt{f * \nu_h} - \sqrt{f * \eta_h}}{h^r} \rightarrow \frac{(-1)^r \Delta_r}{2r!} \frac{f^{(r)}}{\sqrt{f}}, \quad (2.6)$$

where the right side is zero outside $(-S, S)$. The statement includes $\Delta_r = 0$, in which case (2.5) is a little-oh assertion.

For the statistical model, fix $m \geq 1$ and set

$$\mathcal{A}_m = \{a \in [0, 1]^m : a_1 \leq \dots \leq a_m\}, \quad X_{a,v} = B + \sum_{i=1}^m a_i V_i + \sqrt{v} G, \quad (2.7)$$

where the V_i are fresh uniforms on $[-1, 1]$, G is standard normal, and all variables are independent. Let $P_{a,v} = \text{Law}(X_{a,v})$. Zeros pad the factor list to a fixed capacity; they are not observable positive factors. The background law is known.

Theorem 2.2 (Sharp estimation, including the zero-noise boundary). *For n independent observations, define minimax expected loss using $\|\hat{a} - a\|_\infty$ for scales and $|\hat{v} - v|$ for variance. With known fixed $v_0 \geq 0$,*

$$\inf_{\hat{a}} \sup_{a \in \mathcal{A}_m} \mathbb{E}_{a,v_0} \|\hat{a} - a\|_\infty \asymp n^{-1/(4m)}. \quad (2.8)$$

With unknown $v \in [0, V]$, for fixed $V > 0$,

$$\inf_{\hat{a}} \sup_{a \in \mathcal{A}_m, 0 \leq v \leq V} \mathbb{E}_{a,v} \|\hat{a} - a\|_\infty \asymp n^{-1/(4m+4)}, \quad (2.9)$$

$$\inf_{\hat{v}} \sup_{a \in \mathcal{A}_m, 0 \leq v \leq V} \mathbb{E}_{a,v} |\hat{v} - v| \asymp n^{-1/(2m+2)}. \quad (2.10)$$

The constants may depend on m , the fixed background, and v_0 or V . The lower bounds in (2.9)–(2.10) hold already on every fixed relative neighborhood of $(a, v) = (0, 0)$, using pairs for which one variance is exactly zero. In (2.8), hard pairs approach $a = 0$, also when $v_0 = 0$.

The last clause is a local *minimax* statement. It does not assert that the risk of every estimator at the single parameter $(0, 0)$ has a lower bound: the identically zero estimator has zero loss at that one point.

Theorem 2.3 (Detection thresholds). *Test no positive factors, $a = 0$, against $a_m \geq h$, with fixed capacity m . With known $v_0 \geq 0$, the critical separation order is $n^{-1/4}$: consistent testing is possible if $nh^4 \rightarrow \infty$, and the optimal sum of worst-case type-I and type-II errors tends to one if $nh^4 \rightarrow 0$. With unknown $v \in [0, V]$, replace nh^4 by nh^8 ; the critical order is $n^{-1/8}$. Both assertions include alternatives having exactly zero Gaussian variance.*

3 Uniform prefixes and higher-order information

3.1 Smoothness, support, and the endpoint polynomial

Let $A_N = \sum_{j \leq N} b_j$ and $s_N = S - A_N$. Repeated integration of interval indicators shows that f_N is a compactly supported spline of degree $N - 1$, is positive on $(-A_N, A_N)$, and belongs to C^{N-2} for $N \geq 2$. These claims can also be read directly from the formula

$$f_N(x) = \frac{1}{(N-1)! \prod_{j=1}^N 2b_j} \sum_{E \subseteq \{1, \dots, N\}} (-1)^{|E|} \left(x + A_N - 2 \sum_{j \in E} b_j \right)_+^{N-1}. \quad (3.1)$$

For $N = 1$ the power zero means an interval step, with its value at a knot irrelevant. The formula follows inductively from convolution with $(2b_N)^{-1} \mathbf{1}_{[-b_N, b_N]}$. Positivity follows by integrating a positive previous density over the nonempty overlap of the two support intervals.

For later use the endpoint has the particularly simple expression

$$f_N(A_N - d) = f_N(-A_N + d) = C_N d^{N-1}, \quad C_N = \frac{1}{(N-1)! \prod_{j=1}^N 2b_j}, \quad 0 < d < 2 \min_{j \leq N} b_j. \quad (3.2)$$

Indeed, the deficits $b_j - b_j U_j$ are uniform on $[0, 2b_j]$; at such small total deficit their convolution is the unconstrained simplex-volume formula.

Lemma 3.1 (Regularity of the infinite background). *The law in (2.1) has a C^∞ density f , supported on $[-S, S]$, positive on $(-S, S)$. For every fixed $k \geq 0$, $f_N^{(k)} \rightarrow f^{(k)}$ uniformly once N is large enough for these classical derivatives to exist. All derivatives of f vanish at both endpoints.*

Proof. Fix $L \geq k + 2$. Convolution of f_L with the law of the remaining infinite sum gives a density with bounded continuous derivatives through order k . This represents the same law for each k , so gives $f \in C^\infty$. Couple the finite and infinite tails after L with the same uniforms. Their difference is bounded by s_N . Uniform continuity of $f_L^{(k)}$ therefore bounds $\|f_N^{(k)} - f^{(k)}\|_\infty$ by its modulus of continuity at s_N , which tends to zero.

For $|x| < S$, choose N so large that $|x| + s_N < A_N$. Every shift of x by an element of the tail support is then inside the positive interval of f_N . Thus $f(x) = \mathbb{E} f_N(x - (B - B_N)) > 0$. Support inclusion follows directly from $|B| \leq S$; positivity proves equality of the support. Smoothness and identical vanishing outside the support force every endpoint derivative to vanish. $\square$

3.2 A conditional-score inequality

For integer $r \geq 1$ and real $q \geq 2$, put

$$J_{r,q}(g) = \int_{g>0} \left| \frac{g^{(r)}}{g} \right|^q g, \quad J_r(g) = J_{r,2}(g). \quad (3.3)$$

We apply this only to densities whose indicated weak derivative is an integrable function and is zero almost everywhere on their zero set. All sufficiently long prefixes satisfy these conditions.

Lemma 3.2 (Convolution contraction). *For such a density g and any probability law μ ,*

$$J_{r,q}(g * \mu) \leq J_{r,q}(g) \quad (3.4)$$

whenever the right side is finite.

Proof. Write $X \sim g$, $Y \sim \mu$, independently. Differentiating the convolution in the weak sense gives $(g * \mu)^{(r)} = g^{(r)} * \mu$. The ratio of this identity to $g * \mu$ says, for almost every z of positive density,

$$\frac{(g * \mu)^{(r)}(z)}{(g * \mu)(z)} = \mathbb{E} \left[\frac{g^{(r)}(X)}{g(X)} \middle| X + Y = z \right].$$

Conditional Jensen followed by expectation proves the claim. $\square$

This is the standard score-contraction mechanism; its use here does not claim a new general information inequality.

Proposition 3.3 (All-order finiteness and convergence of prefixes). *For every $r \geq 1$, $q \geq 2$, $J_{r,q}(f) < \infty$. Moreover,*

$$J_r(f_N) \downarrow J_r(f) \quad \text{as } N \rightarrow \infty, \quad (3.5)$$

where the sequence is considered from any index $N > 2r$ onward. The limit is strictly positive.

Proof. If $N > rq$, the endpoint formula gives

$$\left| \frac{f_N^{(r)}(A_N - d)}{f_N(A_N - d)} \right|^q f_N(A_N - d) = O(d^{N-1-rq}).$$

This is integrable; away from the endpoints the density is bounded below and its indicated derivatives are bounded. Thus $J_{r,q}(f_N) < \infty$. Convolution with the infinite tail and [theorem 3.2](#) prove finiteness for f . Convolution with one further uniform proves monotonicity of $J_r(f_N)$.

To identify its limit, fix $q > 2$ and a prefix $L > rq$. For all $N \geq L$, $J_{r,q}(f_N) \leq K := J_{r,q}(f_L)$. On every compact interval inside $(-S, S)$ the integrands defining $J_r(f_N)$ converge uniformly to that of $J_r(f)$, by [theorem 3.1](#). On the complementary endpoint region $E_\delta = \{x : |x| > S - \delta\}$, Hölder's inequality gives

$$\int_{E_\delta} \frac{(f_N^{(r)})^2}{f_N} \leq K^{2/q} \mathbb{P}(|B_N| > S - \delta)^{1-2/q}. \quad (3.6)$$

When $s_N < \delta$, the probability is at most $\mathbb{P}(|B| > S - 2\delta)$ by the deterministic coupling bound $|B_N - B| \leq s_N$. This tends to zero as $\delta \downarrow 0$, since B has no endpoint atoms. The same estimate applies to f . Interior convergence and these tail estimates prove (3.5).

If $J_r(f) = 0$, then $f^{(r)}$ is zero throughout $(-S, S)$ by continuity. It is zero outside as well, so f is a polynomial of degree at most $r - 1$ on $\mathbb{R}$. A compactly supported polynomial is zero, contradicting integral one. Hence $J_r(f) > 0$. $\square$

4 The finite-prefix Hellinger expansion

The following lemma isolates the analytic issue. It deliberately requires more uniform factors than twice the first potentially unmatched moment order; [section 6](#) will show why the threshold is sharp.

Lemma 4.1 (A sufficiently long finite prefix). *Let g be the density of a sum of M independent positive-width centered uniforms, with $M > 2r$. If all-moment laws ν, η agree in moments $0, \dots, r-1$, then*

$$H^2(g * \nu_h, g * \eta_h) = \frac{\Delta_r^2}{4(r!)^2} J_r(g) h^{2r} + o(h^{2r}). \quad (4.1)$$

Proof. Write $[-A, A]$ for the support. First suppose the laws are supported on $[-L, L]$, and let $p_h = g * \nu_h$, $q_h = g * \eta_h$. On a fixed compact subinterval of $(-A, A)$, Taylor expansion and moment cancellation give

$$h^{-r}(p_h - q_h) \longrightarrow \frac{(-1)^r \Delta_r}{r!} g^{(r)}. \quad (4.2)$$

The available smoothness is sufficient since $M > 2r$ implies $g \in C^r$. Both densities tend to g , giving the claimed pointwise square-root limit.

Consider a small endpoint collar and write $d = A - |x|$. On the interior part $d \geq 2hL$, all shifted arguments stay inside the polynomial endpoint collar provided its fixed width is reduced once. There the endpoint formula and Taylor's integral remainder yield

$$p_h(x) + q_h(x) \geq cd^{M-1}, \quad |p_h(x) - q_h(x)| \leq Ch^r d^{M-1-r}. \quad (4.3)$$

Constants may depend on the fixed laws and prefix. Consequently

$$h^{-2r}(\sqrt{p_h} - \sqrt{q_h})^2 \leq Cd^{M-1-2r}. \quad (4.4)$$

The exponent is greater than -1 . On the remaining fixed interior the same quantity is bounded, because g is bounded below and its r th derivative is bounded. Dominated convergence is therefore valid away from the shrinking boundary band. That band, including the new support outside $[-A, A]$, has width $O(h)$ and total $p_h + q_h$ mass $O(h^M)$. Its Hellinger contribution is at most that mass, and $h^M = o(h^{2r})$.

We give the extension to unbounded all-moment laws explicitly. Choose $\varepsilon > 0$ with $M(1-\varepsilon) > 2r$ and put $L_h = h^{-\varepsilon}$. Condition each law on $|Z| \leq L_h$ to obtain $\nu^{[h]}, \eta^{[h]}$. For every prescribed K , the discarded mass and each of the finitely many discarded moments through order r are $O(h^K)$: apply Markov's inequality with a sufficiently high absolute moment. The same holds for the effect of normalization. In particular, their moment discrepancies of orders $j < r$ are $O(h^K)$, and their discrepancy at order r tends to Δ_r .

Replacing a law by this conditioning changes the Hellinger distance after any convolution by $O(h^{K/2})$. To see this, couple the original variable and its conditional version so that they differ with probability at most the discarded mass, and use $H^2 \leq 2 \text{TV}$. Choose $K > 2r$, so the change in the Hellinger distance is $o(h^r)$.

For the conditioned densities use the band of width $2hL_h$. Its mass is $O((hL_h)^M) = o(h^{2r})$. On the remaining endpoint collar the term coming from the r th remainder is still bounded as in (4.4), with a uniform constant: the bound uses the uniformly bounded r th absolute moments, not L_h^r . A residual moment of order $j < r$ contributes at most $Ch^{j+K}d^{M-1-j}$ to the density difference. After division by a square-root lower bound and by h^r , its L^2 norm is at most

$$Ch^{j+K-r} \left(\int_0^{d_0} d^{M-1-2j} dd \right)^{1/2} = o(1).$$

The fixed interior has the same bound without the endpoint weight. Thus the residual moments do not affect the limiting norm. The r th term is handled by dominated convergence exactly as before. Finally the triangle inequality for H , and its $o(h^r)$ conditioning error, transfer the expansion to the original laws. $\square$

The proof does not suppose that p_h/g is bounded everywhere. Such a supposition would fail even for a single added uniform, because its support extends beyond that of g .

5 Passing from prefixes to the infinite convolution

5.1 A lower bound and a matching upper bound

Proof of theorem 2.1. Finiteness and positivity of $J_r(f)$ are theorem 3.3. Smoothness, bounded derivatives, and the $(r + 1)$ st absolute moments give, at every fixed x ,

$$(f * \nu_h - f * \eta_h)(x) = \frac{(-1)^r \Delta_r}{r!} h^r f^{(r)}(x) + O(h^{r+1}).$$

For $|x| < S$, division by $\sqrt{f * \nu_h(x)} + \sqrt{f * \eta_h(x)} \rightarrow 2\sqrt{f(x)}$ and Fatou's lemma imply

$$\liminf_{h \downarrow 0} h^{-2r} H^2(f * \nu_h, f * \eta_h) \geq \frac{\Delta_r^2}{4(r!)^2} J_r(f). \quad (5.1)$$

For the upper bound fix $N > 2r$ and let τ_N be the law of the infinite tail after N , so $f = f_N * \tau_N$. Hellinger distance contracts under a common convolution:

$$H^2(f * \nu_h, f * \eta_h) \leq H^2(f_N * \nu_h, f_N * \eta_h). \quad (5.2)$$

Indeed, pointwise Cauchy–Schwarz gives $\sqrt{(p * \tau)(q * \tau)} \geq \sqrt{pq} * \tau$, and integration shows that affinity cannot decrease. Applying theorem 4.1 gives

$$\limsup_{h \downarrow 0} h^{-2r} H^2(f * \nu_h, f * \eta_h) \leq \frac{\Delta_r^2}{4(r!)^2} J_r(f_N).$$

Now send N to infinity and apply (3.5). This matches (5.1) and proves the exact norm limit.

For completeness the stronger L^2 conclusion follows too. The scaled square-root differences converge pointwise inside the support to the stated function. Outside it, for any fixed positive distance d from the support,

$$(f * \nu_h)(x) \leq \|f\|_\infty \mathbb{P}(|Z| \geq d/h) = O(h^K)$$

for every K , and similarly for η . Thus the scaled functions converge almost everywhere on $\mathbb{R}$. Their L^2 norms are bounded and converge to the norm of the pointwise limit. Boundedness and almost-everywhere convergence imply weak L^2 convergence: first test against bounded functions of compact support, using uniform integrability in L^1 on that support, then approximate an arbitrary L^2 test function. Weak convergence plus convergence of norms in a Hilbert space gives strong convergence, proving (2.6). $\square$

The order of limits matters. We first take $h \downarrow 0$ at a fixed finite prefix, then send the prefix length to infinity. No uniform estimate in both variables has been silently assumed.

5.2 What happens to the new support?

Proposition 5.1 (Leakage is smaller than every power). *For any all-moment perturbation Z , independent of B , and every integer $K \geq 1$,*

$$\mathbb{P}(|B + hZ| > S) = O(h^K). \quad (5.3)$$

*Nevertheless, for a nonzero uniform perturbation, $\chi^2(f * \text{Law}(hU), f) = \infty$ for every $h > 0$.*

Proof. If $B > S - d$, then $B_N > A_N - d$, since the tail is at most s_N . The finite-prefix simplex bound gives $\mathbb{P}(B_N > A_N - d) \leq C'_N d^N$ for small d , and increasing the constant makes this valid for every $d \geq 0$. Reflection gives the other endpoint. Conditioning on Z therefore bounds the leakage by $C'_N h^N \mathbb{E}|Z|^N$. Set $N = K$.

For an added uniform the convolution is positive on $(-S - h, S + h)$, including sets outside $[-S, S]$ of positive measure, while f is zero there. It is not absolutely continuous relative to the law with density f , so the indicated chi-square divergence is infinite. $\square$

Small leakage is not zero leakage. The exact Hellinger theorem is therefore not a disguised finite-chi-square argument.

5.3 An identifiable, beyond-all-powers consequence

Corollary 5.2 (All moments may hide every algebraic scale). *If distinct all-moment laws ν, η have the same moments of every nonnegative integer order, then, for every $K \geq 1$,*

$$H(f * \nu_h, f * \eta_h) = o(h^K). \quad (5.4)$$

The two convolved laws are nonetheless distinct for every $h > 0$.

Proof. Apply [theorem 2.1](#) with $r = K$ and $\Delta_r = 0$. For distinctness, the characteristic function of the compactly supported background extends to an entire function and equals one at zero. Its real zeros are therefore isolated. Equality of the convolved characteristic functions would force those of ν_h, η_h to agree off that discrete zero set, and then everywhere by continuity. Uniqueness of characteristic functions would give $\nu = \eta$, a contradiction. $\square$

An explicit standard moment-indeterminate construction uses the lognormal density

$$\ell(z) = \frac{e^{-(\log z)^2/2}}{z\sqrt{2\pi}}, \quad z > 0, \quad \ell_\theta(z) = \ell(z)(1 + \theta \sin(2\pi \log z)), \quad 0 < |\theta| < 1.$$

Both are positive probability densities with all moments. For Y standard normal and every integer $k \geq 0$,

$$\mathbb{E}[e^{kY} \sin(2\pi Y)] = \text{Im } e^{(k+2\pi i)^2/2} = e^{k^2/2-2\pi^2} \sin(2\pi k) = 0.$$

This verifies the construction directly; moment-indeterminate lognormal families are classical, with further context in [5]. At any amplitude $h_n = n^{-\alpha}$, $\alpha > 0$, the two n -sample laws have total variation tending to zero, by (5.4) and $H^2(P^n, Q^n) \leq nH^2(P, Q)$. This does not contradict the finite-uniform-factor rates: these lognormal perturbations are outside that finite-dimensional model.

6 A sharp finite-boundary phase diagram

An infinite uniform convolution has arbitrarily long finite prefixes. A finite convolution does not. The resulting distinction can be quantified exactly, including the borderline logarithm.

Let g_M be the density of $\sum_{i=1}^M b_i U_i$, now with M fixed, and let $A = \sum_{i=1}^M b_i$ and C_M be as in (3.2). Suppose ν, η are symmetric all-moment laws whose first unmatched moment has even order $r \geq 2$, so $\Delta_r \neq 0$. Define

$$\Psi_{\nu,M}(t) = \int (z-t)_+^{M-1} d\nu(z), \quad D_M(t) = \sqrt{\Psi_{\nu,M}(t)} - \sqrt{\Psi_{\eta,M}(t)}. \quad (6.1)$$

For $M = 1$, $(z-t)_+^0$ means $\mathbf{1}_{\{z>t\}}$.

Theorem 6.1 (Finite-uniform boundary trichotomy). *Under these hypotheses, as $h \downarrow 0$:*

$$M < 2r : \quad H^2(g_M * \nu_h, g_M * \eta_h) \sim 2C_M A_M h^M, \quad (6.2)$$

$$M = 2r : \quad H^2(g_M * \nu_h, g_M * \eta_h) \sim \frac{C_M \Delta_r^2}{2} \binom{2r-1}{r} h^{2r} \log(1/h), \quad (6.3)$$

$$M > 2r : \quad H^2(g_M * \nu_h, g_M * \eta_h) \sim \frac{\Delta_r^2}{4(r!)^2} J_r(g_M) h^{2r}. \quad (6.4)$$

Here $A_M = \int_{\mathbb{R}} D_M(t)^2 dt \in (0, \infty)$ when $M < 2r$. In that case the leading constant depends on the full endpoint profiles, not merely on Δ_r .

Proof. The third case is [theorem 4.1](#). We prove the other two and justify that no interior spline knot contributes at leading order.

Endpoint reduction. Choose a fixed collar width $d_0 > 0$ small enough that the two endpoint collars are disjoint and (3.2) holds on a collar twice as wide. At the right edge write $x = A + ht$. Replacing $g_M(y)$ by $C_M(A - y)_+^{M-1}$ gives exactly

$$(g_M * \nu_h)(A + ht) = C_M h^{M-1} \Psi_{\nu, M}(t)$$

when the shifted argument is in that polynomial collar. The error for $|x - A| \leq d_0$ only involves $|Z| \geq d_0/h$, after reducing d_0 if necessary. All absolute moments imply that its L^1 norm on the collar is $O(h^K)$ for every prescribed K ; the extra polynomial factor of degree $M - 1$ is absorbed by taking a higher moment. The same statement holds for η . Since $\|\sqrt{p} - \sqrt{q}\|_2^2 \leq \|p - q\|_1$ for nonnegative functions, these errors are negligible to any prescribed algebraic order in the Hellinger calculation.

Thus the right collar contributes, to the required accuracy,

$$C_M h^M \int_{-d_0/h}^{d_0/h} D_M(t)^2 dt. \quad (6.5)$$

The left collar has the same value by symmetry. Outside both the support and the collars, the total density mass is smaller than every power of h , again by a high-moment tail estimate.

Asymptotics of the profiles. Put $k = M - 1$ and let $u \rightarrow \infty$. Replacing $(u + Z)_+^k$ by $(u + Z)^k$ creates an error $O(u^{-K})$ for every prescribed K , by the all-moment assumption. If $k \geq r$, binomial expansion and matching of moments below r give

$$D_M(-u) \sim \frac{\Delta_r}{2} \binom{M-1}{r} u^{(M-1)/2-r}. \quad (6.6)$$

If $k < r$, all polynomial terms cancel and $D_M(-u)$ decays faster than any inverse power, after division by its denominator of order $u^{k/2}$. For $t \rightarrow +\infty$, each profile itself decays faster than any inverse power, since $\mathbb{E}(Z - t)_+^k \leq t^{-K} \mathbb{E}|Z|^{k+K}$ for $t > 0$.

Consequently D_M^2 is integrable when $M < 2r$. Its integral is positive: if the profiles agreed almost everywhere, their M th distributional derivatives would agree, whereas

$$\frac{d^M}{dt^M} \Psi_{\nu, M} = (-1)^M (M-1)! \nu$$

as distributions. This would imply $\nu = \eta$, contrary to $\Delta_r \neq 0$. This proves the constant in (6.2). When $M = 2r$, (6.6) instead gives

$$D_M(-u)^2 \sim \frac{\Delta_r^2}{4} \binom{2r-1}{r}^2 \frac{1}{u}.$$

Integrating to d_0/h in (6.5), and adding the two endpoints, gives (6.3).

The interior. On a compact subinterval of $(-A, A)$ the two convolved densities are bounded below for small h . If $M \geq r + 1$, the spline belongs to $W^{r, \infty}$; Taylor's integral bound and moment cancellation give a uniform density difference $O(h^r)$ there. Its Hellinger contribution is $O(h^{2r})$, which is negligible in the first two regimes.

If $2 \leq M \leq r$, use the finite truncated-power representation (3.1). Since all moments through degree $M - 1$ match, the difference produced by a single knot τ is h^{M-1} times a profile in $(x - \tau)/h$ that decays faster than any inverse power in both directions. Its squared L^2 integral is $O(h^{2M-1})$.

There are finitely many knots, even when subset sums coincide; the squared norm of their sum has the same order up to a fixed constant. Division by the interior density lower bound preserves this estimate. As $2M - 1 > M$ for $M \geq 2$, it is negligible compared with h^M . For $M = 1$ there are no interior knots and the density is constant in the interior; only high-moment tails can affect that fixed interior region. This completes the proof. $\square$

Corollary 6.2 (Simple testing scales for a finite prefix). *For the simple pair in [theorem 6.1](#), the critical amplitude orders are*

$$\begin{cases} n^{-1/M}, & M < 2r, \\ (n \log n)^{-1/(2r)}, & M = 2r, \\ n^{-1/(2r)}, & M > 2r. \end{cases} \quad (6.7)$$

Precisely, the product affinity tends to one when $nH^2 \rightarrow 0$, to zero when $nH^2 \rightarrow \infty$, and to $e^{-\lambda/2}$ when $nH^2 \rightarrow \lambda$.

Proof. Independence gives $\rho(P^n, Q^n) = (1 - H^2(P, Q)/2)^n$. Insert the three asymptotics. The inequalities $\text{TV} \leq H$ and $1 - \text{TV} \leq \rho$ identify the indistinguishable and distinguishable regimes. $\square$

These are simple-pair testing scales. They are not a blanket minimax theorem for all finite-prefix parameter models. In particular, boundary information can improve a specific testing problem even though moment fitting alone does not use that information.

7 Two explicit boundary transitions

First compare $\nu = \text{Law}(U)$ with $\eta = \delta_0$. The first unmatched moment is $r = 2$, with $\Delta_2 = 1/3$. For M background uniforms all of half-length one,

$$C_M = \frac{1}{2^M(M-1)!}. \quad (7.1)$$

At $M = 1$, direct integration of the two linear edge ramps gives, for $0 < h < 1$,

$$H^2(g_1 * \text{Law}(hU), g_1) = \frac{2(\sqrt{2} - 1)}{3} h. \quad (7.2)$$

At the critical number $M = 4$,

$$H^2(g_4 * \text{Law}(hU), g_4) \sim \frac{1}{192} h^4 \log(1/h). \quad (7.3)$$

For every $M > 4$, the asymptotic is $J_2(g_M)h^4/144$. For the infinite up density, it is likewise

$$H^2(\text{up} * \text{Law}(hU), \text{up}) \sim \frac{J_2(\text{up})}{144} h^4. \quad (7.4)$$

Thus a nonzero added support does not force an h -scale Hellinger contribution: the order of boundary vanishing controls its importance.

Second compare a uniform perturbation with a variance-matched Gaussian: $\nu = \text{Law}(U)$ and $\eta = N(0, 1/3)$. The first unmatched moment is $r = 4$,

$$\Delta_4 = \frac{1}{5} - \frac{1}{3} = -\frac{2}{15}. \quad (7.5)$$

The critical finite prefix now has $M = 8$ factors, and

$$H^2(g_8 * \text{Law}(hU), g_8 * N(0, h^2/3)) \sim \frac{7}{829440} h^8 \log(1/h). \quad (7.6)$$

The infinite up-law counterpart is

$$H^2(\text{up} * \text{Law}(hU), \text{up} * N(0, h^2/3)) \sim \frac{J_4(\text{up})}{129600} h^8. \quad (7.7)$$

These constants use the normalization (2.3). Halving the definition of squared Hellinger distance would halve every displayed Hellinger constant, but would not change any rate exponent.

8 The finite algebra needed for recovery

8.1 Cumulants of a uniform factor

Let $p_k(x) = \sum_{i=1}^m x_i^k$, with $x_i = a_i^2$. Cumulants are coefficients in $\log \mathbb{E}e^{tX} = \sum_{j \geq 1} \kappa_j(X) t^j / j!$ near zero. For U uniform on $[-1, 1]$,

$$c_k := \kappa_{2k}(U) = \frac{2^{2k} B_{2k}}{2k} = \frac{(-1)^{k+1} (2k)! \zeta(2k)}{k \pi^{2k}} \neq 0. \quad (8.1)$$

For example, $c_1 = 1/3$, $c_2 = -2/15$, $c_3 = 16/63$. One verification is to take logarithms in $\sinh t/t = \prod_{j \geq 1} (1 + t^2/(\pi^2 j^2))$ for $|t| < \pi$ and compare coefficients. Cumulant additivity and scaling give

$$\kappa_2(X_{a,v}) - \kappa_2(B) = v + \frac{1}{3} p_1(x), \quad (8.2)$$

$$\kappa_{2k}(X_{a,v}) - \kappa_{2k}(B) = c_k p_k(x), \quad k \geq 2. \quad (8.3)$$

Known variance reveals $p_1, \dots, p_m$ through cumulant order $2m$. Unknown variance conceals p_1 and reveals the consecutive window $p_2, \dots, p_{m+1}$ through order $2m + 2$.

For the background (2.2), the explicit formulas are

$$\kappa_{2k}(B) = \frac{c_k}{2^{2k} - 1}, \quad \text{Var}(B) = \frac{1}{9}, \quad \kappa_4(B) = -\frac{2}{225}. \quad (8.4)$$

More generally replace $(2^{2k} - 1)^{-1}$ by $\sum_j b_j^{2k}$. Absolute summability justifies differentiation of the logarithmic moment product in a neighborhood of zero.

8.2 A credited shifted-power-sum inequality

The following result and its counting-function proof are reproduced from [7, Theorem 4.1]. They are not counted as a new theorem of this article.

Lemma 8.1 (Consecutive shifted power sums). *For ordered $x, y \in [0, 1]^m$, integer $s \geq 0$, and*

$$E_s(x, y) = \max_{s+1 \leq k \leq s+m} |p_k(x) - p_k(y)|,$$

one has

$$\|x - y\|_\infty \leq 8E_s(x, y)^{1/(m+s)}, \quad \|\sqrt{x} - \sqrt{y}\|_\infty \leq \sqrt{8}E_s(x, y)^{1/(2m+2s)}. \quad (8.5)$$

Proof. Define $C(t) = \#\{i : x_i \leq t\} - \#\{i : y_i \leq t\}$ on $(0, 1)$. The nonzero sign of this integer-valued step function changes at most $m - 1$ times. Indeed, resolve its jumps into m unit up-steps and m unit down-steps. The resulting walk starts and ends at zero and has at most m excursions; combining consecutive excursions of the same sign can only reduce the number of sign blocks. Intermediate steps at a common atom have zero interval length and create no extra sign block of C .

Choose $z_1, \dots, z_q$ between opposite-sign blocks, $q \leq m - 1$, and choose the sign of $Q(t) = \pm \prod_{j=1}^q (t - z_j)$ so that $QC \geq 0$ almost everywhere. The sum of the absolute monomial coefficients of Q is at most 2^q .

Put $d = \|x - y\|_\infty > 0$. Interchanging the vectors if necessary, some $y_i - x_i = d$. On (x_i, y_i) , $C \geq 1$, so no sign-change point lies in that interval. On its middle half, $t \geq d/4$ and $|t - z_j| \geq d/4$. Hence

$$\int_0^1 t^s Q(t) C(t) dt \geq \frac{d}{2} \left(\frac{d}{4}\right)^{s+q}. \quad (8.6)$$

Direct integration of the counting functions gives

$$p_k(x) - p_k(y) = -k \int_0^1 t^{k-1} C(t) dt.$$

Expanding Q therefore bounds the left side of (8.6) above by $2^q E_s / (s+1)$. Since $d \leq 1$ and $q \leq m-1$,

$$d^{m+s} \leq \frac{2^{2s+3m-2}}{s+1} E_s \leq 8^{m+s} E_s.$$

This proves the first inequality; the second follows from $|\sqrt{u} - \sqrt{v}| \leq \sqrt{|u - v|}$ for nonnegative u, v . $\square$

Equal unit multiplicities and nonnegative nodes are essential hypotheses of this proof. Arbitrary signed or unequal-weight moment inversion is a different problem.

8.3 Exact moment-preserving pairs

The construction below, also used in [7], supplies fixed algebraic shapes. Our new analytic theorem will determine the Hellinger distance between their scaled statistical realizations.

Lemma 8.2 (Hard shapes). *For every $m \geq 1$ there exist distinct positive ordered vectors u, w in $(0, 1)^m$ such that*

$$p_1(u) = p_1(w), \dots, p_{m-1}(u) = p_{m-1}(w), \quad p_m(u) \neq p_m(w). \quad (8.7)$$

There also exist such vectors with

$$\begin{aligned} p_2(u) &= p_2(w), \dots, p_m(u) = p_m(w), \\ p_1(u) &\neq p_1(w), \quad p_{m+1}(u) \neq p_{m+1}(w). \end{aligned} \quad (8.8)$$

Empty strings of equalities, when $m = 1$, impose no constraint.

Proof. Start at distinct $0 < u_1 < \dots < u_m < 1$, let $P(z) = \prod_i (z - u_i)$, and define the tangent

$$\beta_i = \frac{1}{u_i^s P'(u_i)}, \quad s \in \{0, 1\}.$$

Lagrange interpolation of monomials of degrees at most $m-1$ gives

$$\sum_i \frac{u_i^j}{P'(u_i)} = \begin{cases} 0, & 0 \leq j < m-1, \\ 1, & j = m-1. \end{cases}$$

Thus the differentials of $p_{s+1}, \dots, p_{s+m-1}$ annihilate β , whereas $Dp_{s+m}(u)\beta = s+m$. The constraint differentials have rank $m-1$, by the rectangular Vandermonde matrix after scaling nonzero rows and columns. The implicit function theorem gives a local one-dimensional curve preserving those moments, with tangent β ; another sufficiently nearby point gives the required change in the final moment. The coordinates stay distinct, positive, and less than one.

For $s = 1$, partial fractions of $1/P$ evaluated at zero also give

$$Dp_1(u)\beta = \sum_i \frac{1}{u_i P'(u_i)} = \frac{(-1)^{m+1}}{\prod_i u_i} \neq 0.$$

Choose the second point close enough that both this moment and p_{m+1} change. This proves the additional assertion in (8.8). $\square$

9 A feasible estimator and the upper bounds

Proposition 9.1 (Finite-grid moment fitting). *There are measurable estimators attaining the upper bounds in [theorem 2.2](#), uniformly over the stated parameter sets.*

Proof. For each required raw moment order j form the sample average of X^j . Clip it to a fixed interval containing every true raw moment of that order in the model. All absolute moments of $X_{a,v}$ of any fixed order are uniformly bounded: the background and uniform factors are bounded, and the Gaussian variance is bounded. Consequently the clipped raw-moment estimates have uniformly bounded mean squared error C_j/n .

Cumulants through any fixed order are polynomials in raw moments. On the fixed clipping box these polynomials are Lipschitz. Hence the resulting cumulant estimates satisfy

$$\mathbb{E}|\hat{\kappa}_j - \kappa_j(X_{a,v})|^2 \leq C'_j/n. \quad (9.1)$$

With known variance, use (8.2)–(8.3) to estimate $p_1, \dots, p_m$. With unknown variance, estimate $p_2, \dots, p_{m+1}$. Write $s = 0$ or $s = 1$ respectively, and $\hat{p}_k$ for these estimates. Their maximum error D_n has $\mathbb{E}D_n \leq Cn^{-1/2}$.

On the finite ordered grid

$$\mathcal{X}_{m,n} = \{x \in \{0, 1/n, \dots, 1\}^m : x_1 \leq \dots \leq x_m\},$$

minimize $\max_{s+1 \leq k \leq s+m} |p_k(x) - \hat{p}_k|$, breaking ties lexicographically. The true squared vector has an ordered rounded grid point within $1/n$ coordinatewise; its power-sum discrepancy is at most C/n . Thus the chosen $\hat{x}$ satisfies

$$E_s(\hat{x}, x) \leq 2D_n + C/n. \quad (9.2)$$

Set $\hat{a}_i = \sqrt{\hat{x}_i}$. Apply [theorem 8.1](#), then Jensen's inequality for the concave positive powers involved, to obtain

$$\mathbb{E} \|\hat{x} - x\|_\infty \leq Cn^{-1/(2m+2s)}, \quad (9.3)$$

$$\mathbb{E} \|\hat{a} - a\|_\infty \leq Cn^{-1/(4m+4s)}. \quad (9.4)$$

In the unknown-variance case define

$$\hat{v} = \Pi_{[0,V]} \left(\hat{\kappa}_2 - \kappa_2(B) - \frac{1}{3}p_1(\hat{x}) \right),$$

where Π is clipping to the interval. Since clipping cannot increase distance to the true v ,

$$\mathbb{E}|\hat{v} - v| \leq Cn^{-1/2} + \frac{m}{3} \mathbb{E} \|\hat{x} - x\|_\infty \leq C'n^{-1/(2m+2)}.$$

All minimizations were over finite fixed sets, so measurability is explicit. $\square$

The grid has $\binom{n+m}{m}$ points. This is a proof of an attainable statistical rate for fixed m , not a proposed scalable implementation. Exact data also determine the parameters uniquely by [theorem 8.1](#) and the variance identity; no density inversion or use of empirical total variation is needed.

10 Lower bounds at vanishing factors and zero variance

10.1 Known variance

Choose u, w from (8.7). Let

$$Z_u = \sum_i \sqrt{u_i} V_i, \quad Z_w = \sum_i \sqrt{w_i} V_i, \quad a(h) = h\sqrt{u}, \quad a'(h) = h\sqrt{w}.$$

Both perturbations are symmetric. Their cumulants through order $2m - 1$ agree, and the difference in their $2m$ th cumulants is $c_m(p_m(u) - p_m(w)) \neq 0$. Raw moments are triangular polynomials in cumulants with coefficient one on the highest cumulant; therefore raw moments agree through order $2m - 1$, and

$$\Delta_{2m} = c_m(p_m(u) - p_m(w)). \quad (10.1)$$

At zero Gaussian variance, [theorem 2.1](#) now gives

$$H^2(P_{a(h),0}, P_{a'(h),0}) \sim \frac{\Delta_{2m}^2}{4((2m)!)^2} J_{2m}(f) h^{4m}. \quad (10.2)$$

At any fixed known $v_0 > 0$, a common Gaussian convolution contracts Hellinger distance, so the same pair has $H^2 = O(h^{4m})$. Its scale separation is a fixed positive multiple of h .

10.2 Unknown variance: one member exactly unsmoothed

Choose u, w from (8.8), and define

$$t_u = \max\{0, (p_1(w) - p_1(u))/3\}, \quad t_w = \max\{0, (p_1(u) - p_1(w))/3\}. \quad (10.3)$$

Exactly one of these numbers is zero and the other is strictly positive. Their purpose is the identity $t_u + p_1(u)/3 = t_w + p_1(w)/3$. Let

$$Z_u = \sum_i \sqrt{u_i} V_i + \sqrt{t_u} G, \quad Z_w = \sum_i \sqrt{w_i} V_i + \sqrt{t_w} G.$$

The second cumulants agree by construction, cumulants $4, 6, \dots, 2m$ agree by the power-sum constraints, and odd cumulants vanish. The first unmatched raw moment is therefore

$$\Delta_{2m+2} = c_{m+1}(p_{m+1}(u) - p_{m+1}(w)) \neq 0. \quad (10.4)$$

Both noise laws are all-moment, so [theorem 2.1](#) applies even though only one contains a Gaussian. With

$$a(h) = h\sqrt{u}, \quad v(h) = h^2 t_u, \quad a'(h) = h\sqrt{w}, \quad v'(h) = h^2 t_w,$$

we obtain

$$H^2(P_{a(h),v(h)}, P_{a'(h),v'(h)}) \sim \frac{\Delta_{2m+2}^2}{4((2m+2)!)^2} J_{2m+2}(f) h^{4m+4}. \quad (10.5)$$

The scale separation is comparable to h and the variance separation to h^2 . All parameters tend to $(0, 0)$, and for sufficiently small h lie in $\mathcal{A}_m \times [0, V]$. One variance is exactly zero at every h .

10.3 The statistical conclusion

Completion of theorem 2.2. For any pair of laws, tensorization yields

$$H^2(P^n, Q^n) = 2 \left[1 - \left(1 - \frac{1}{2} H^2(P, Q) \right)^n \right] \leq n H^2(P, Q). \quad (10.6)$$

If two parameter values have metric separation d , any estimator has maximum expected loss at least

$$\frac{d}{4} (1 - \text{TV}(P^n, Q^n)). \quad (10.7)$$

Indeed, deciding which parameter is nearer to the estimate incurs loss at least $d/2$ whenever the resulting test errs; the sum of its two error probabilities is at least $1 - \text{TV}$.

In the known-variance pair choose $h = \gamma n^{-1/(4m)}$, with a fixed sufficiently small $\gamma > 0$. Then (10.2), contraction if needed, and (10.6) give product total variation at most $1/2$ for all sufficiently large n . Equation (10.7) gives the lower bound $cn^{-1/(4m)}$.

For the unknown-variance pair choose $h = \gamma n^{-1/(4m+4)}$. The same argument using (10.5) gives lower bounds of order h for scale loss and h^2 for variance loss. These are exactly (2.9)–(2.10). The pairs eventually lie in every fixed relative neighborhood of $(0, 0)$, proving the local minimax clause. Theorem 9.1 supplied all upper bounds. $\square$

10.4 Sharp deterministic Hellinger inverse exponents

Corollary 10.1 (Inverse moduli). *On the parameter sets in theorem 2.2,*

$$\|a - a'\|_\infty \leq CH(P_{a,v_0}, P_{a',v_0})^{1/(2m)} \quad \text{for known } v_0, \quad (10.8)$$

$$\|a - a'\|_\infty \leq CH(P_{a,v}, P_{a',v'})^{1/(2m+2)}, \quad (10.9)$$

$$|v - v'| \leq CH(P_{a,v}, P_{a',v'})^{1/(m+1)}. \quad (10.10)$$

None of the three exponents can be increased in a uniform inequality, even locally near the corresponding all-zero factor configuration.

Proof. For densities p, q , Cauchy–Schwarz gives

$$\left| \int x^j (p - q) dx \right| \leq H(p, q) (2\mathbb{E}_p |X|^{2j} + 2\mathbb{E}_q |X|^{2j})^{1/2}.$$

The uniform moment bounds and Lipschitz cumulant polynomials therefore control each relevant power-sum discrepancy by CH . Apply theorem 8.1; for variance also use (8.2). The scaled pairs above have parameter gaps of order h or h^2 and Hellinger distances bounded above by order h^{2m} or h^{2m+2} . They exclude every larger exponent. $\square$

11 Detecting a factor is easier than recovering every factor

Let $\mathcal{T}_n(h)$ be the infimum over tests of the sum of their worst-case null and alternative errors, with null $a = 0$ and alternative $a_m \geq h$. In the unknown-variance model the supremum over $v \in [0, V]$ is taken separately under the two hypotheses. This is essential: the null and alternative need not have the same nuisance variance.

Proof of theorem 2.3. With known v_0 , define the estimated variance deficit

$$\hat{D}_2 = \hat{\kappa}_2 - \kappa_2(B) - v_0.$$

Its true value is zero under the null and at least $h^2/3$ under the alternative. Reject when $\hat{D}_2 > h^2/6$. The moment-estimation bound (9.1) and Markov's inequality applied to squared error give

$$\mathcal{T}_n(h) \leq \frac{C}{nh^4}. \quad (11.1)$$

With unknown variance instead use $\hat{D}_4 = \hat{\kappa}_4 - \kappa_4(B)$. Its true value is zero under the null and at most $-2h^4/15$ under the alternative, because $\kappa_4(X) - \kappa_4(B) = -(2/15) \sum_i a_i^4$. Reject when $\hat{D}_4 < -h^4/15$. The same argument gives

$$\mathcal{T}_n(h) \leq \frac{C}{nh^8}. \quad (11.2)$$

For the known-variance lower bound, compare no factor with just one factor of half-length h . At $v_0 = 0$, (7.4) with f in place of u gives Hellinger squared asymptotic $J_2(f)h^4/144$. For $v_0 > 0$, use common-convolution contraction. If $nh^4 \rightarrow 0$, (10.6) implies that product total variation tends to zero, forcing the error sum to be at least $1 - o(1)$.

For the unknown-variance lower bound, use the null with $a = 0, v = h^2/3$ and the alternative with one half-length $h, v = 0$. Both lie in the parameter set for small h . The perturbations have matching variance and first differ in their fourth moment, so

$$H^2(P_{(0,\dots,0,h),0}, P_{0,h^2/3}) \sim \frac{J_4(f)}{129600} h^8. \quad (11.3)$$

If $nh^8 \rightarrow 0$, product total variation again tends to zero. The trivial always-accept test has error sum one, so in both lower-bound regimes the optimal sum tends exactly to one. Together with (11.1)–(11.2), this proves the threshold statements. $\square$

For $m > 1$, the worst-case recovery rate is strictly slower than the corresponding detection threshold. The hard recovery pairs use several coordinated small factors whose low cumulants cancel between the two models. In a test against no factors, all contributions to the first unconfounded even power sum have the same sign. There is no such cancellation to exploit.

12 Concrete two-factor hard pairs

The implicit-function construction proves existence for every capacity. For $m = 2$ the following exact rational shapes make both mechanisms transparent. These examples use squared half-lengths.

12.1 Known variance

Take

$$u = (1/4, 3/4), \quad w = (1/8, 7/8). \quad (12.1)$$

Their first power sums agree and $p_2(u) - p_2(w) = -5/32$. Hence their scaled uniform noises have first unmatched raw moment coefficient $\Delta_4 = (-2/15)(-5/32) = 1/48$. At zero Gaussian variance,

$$H^2(P_{h\sqrt{u},0}, P_{h\sqrt{w},0}) \sim \frac{J_4(f)}{5308416} h^8. \quad (12.2)$$

The two scale vectors differ by a fixed positive multiple of h , giving the $n^{-1/8}$ two-factor recovery obstruction.

12.2 Unknown variance, one exactly zero

Keep $u = (1/4, 3/4)$ and take

$$w = (39/404, 317/404). \quad (12.3)$$

Direct calculation gives

$$p_2(u) = p_2(w) = 5/8, \quad p_1(u) = 1, \quad p_1(w) = 89/101.$$

Thus choose $t_u = 0, t_w = 4/101$. The second cumulants of the two unscaled perturbations are both $1/3$, their fourth cumulants agree, and their sixth raw-moment difference is

$$\Delta_6 = \frac{16}{63}(p_3(u) - p_3(w)) = -\frac{85164}{7212107}. \quad (12.4)$$

Therefore

$$H^2(P_{h\sqrt{u},0}, P_{h\sqrt{w},(4/101)h^2}) \sim \frac{\Delta_6^2}{4(6!)^2} J_6(f) h^{12}. \quad (12.5)$$

This is a zero-noise-boundary realization of the $n^{-1/12}$ scale and $n^{-1/6}$ variance obstructions for capacity two, not an appeal to a positive variance floor.

13 Computational checks and their limits

The package includes `verification.py`, using SymPy 1.14.0 and mpmath 1.3.0. The executed run passed 2,100 exact rational instances of (8.5) for $1 \leq m \leq 7$ and $0 \leq s \leq 2$, using seed 20260929; it also passed 16 exact tangent identities for $1 \leq m \leq 8$, $s = 0, 1$. It verifies the first six uniform cumulants, the displayed hard-pair equalities, and the rational Hellinger constants. These checks independently test algebraic ingredients and transcription; they do not replace the all-parameter proofs.

The numerical part uses 60-decimal arithmetic and piecewise integration at every relevant spline knot. It computes squared Hellinger distances for $g_M * \text{Law}(hU)$ versus g_M , with $M = 1, 3, 4, 5$ and $h = 2^{-3}, \dots, 2^{-7}$. There are twenty diagnostic rows. The finite spline formula is (3.1); convolution with the added uniform is obtained by integrating each truncated power exactly before quadrature. The full data are in `data/hellinger_diagnostics.csv`.

M	Quantity at $h = 1/128$	Computed value	Predicted limit
1	H^2/h	0.276142374915	0.276142374915
3	H^2/h^3	0.017668468430	0.017661491088
4	logarithmic slope $L(h)$	0.005208430349	1/192
5	H^2/h^4	0.007795167122	0.007819937017

Table 1: Finite-prefix diagnostics; numerical quadrature, not interval certificates. Here $L(h) = [H^2(h)/h^4 - H^2(2h)/(2h)^4]/\log 2$.

For $M = 3$, the predicted limit is $2C_3A_3 = A_3/8$ and the independently integrated profile is

$$A_3 \approx 0.141291928707536383107313127009.$$

For $M = 5$ the computation gives

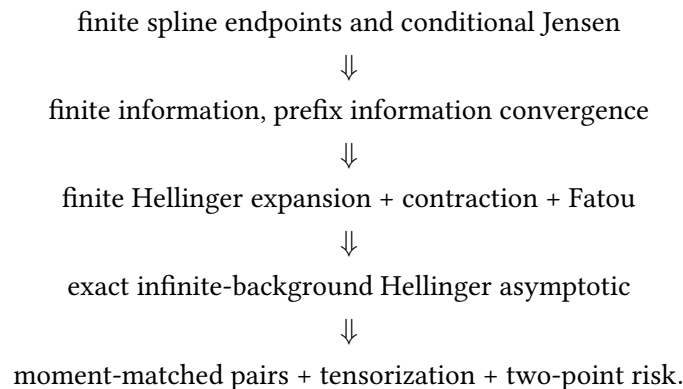
$$J_2(g_5) \approx 1.126070930490470658853603770783, \quad J_2(g_5)/144 \approx 0.00781993701729493513.$$

The critical $M = 4$ ratio $H^2/(h^4 \log(1/h))$ converges slowly because of the next constant term. The slope $L(h)$ removes that leading constant and is a more informative finite- h diagnostic of the coefficient $1/192$. The one-uniform formula (7.2) is also checked numerically against its exact value at every tested amplitude.

No interval enclosure of an information integral is claimed. In particular, none of the displayed finite-prefix numbers is described as a computed $J_r(\text{up})$. The package does not simulate a minimax risk, fit the enormous moment grid, or certify an asymptotic statement by finite sampling. The exact asymptotic constants for the up law remain the rigorously finite, positive integrals in the theorems.

14 Proof audit and formalization route

The dependency chain is short enough to audit structurally:



The statistical upper bounds use a separate branch: uniform moment bounds, finite cumulant polynomials, and the credited inverse inequality.

Three distinctions are essential. First, C^∞ smoothness alone is not used as a theorem that a weighted likelihood expansion must hold. The uniform-prefix representation supplies the missing quantitative control. Second, passing to the infinite convolution uses convergence of information, not just its lower semicontinuity; the higher-score moment bound in (3.6) is what establishes the required direction. Third, no upper bound on chi-square divergence relative to the unsmoothed law is asserted. It is actually infinite in a central example.

A useful proof-assistant implementation would separate four interfaces. The first is the finite uniform-convolution density, support, and endpoint formula. The second is weak differentiation of convolution, conditional Jensen, and weighted uniform integrability. The third is Hellinger affinity, its contraction, tensorization, and the prefix-limit argument. The fourth is finite moment algebra and the statistical decision bound. A certified checker for rational tangent identities could accompany the final interface, but cannot replace the analytic interfaces.

No Lean or Rocq files are delivered as verified proofs. Nothing inherits formal status merely from its association with the ProveIt repository. Likewise, the article does not establish a full local asymptotic normality theorem: the L^2 square-root expansion compares prescribed perturbation curves, and identifying an entire limiting statistical experiment needs additional uniformity.

15 Further research questions and concrete success criteria

Research question 1 (Mixed collision strata at the zero-noise boundary). At a base factor configuration with positive squared-scale multiplicities $m_1, \dots, m_s$ and m_0 zero slots, determine the exact local pairwise Hellinger inverse exponent with unknown variance. The companion manuscript proposes the denominator

$$\max(\{m_1, \dots, m_s\} \cup \{2(m_0 + 1) : m_0 > 0\}).$$

The present paper proves the fully vanishing case, not this entire formula. A solution should supply both a local upper modulus and moment-matched likelihood pairs for every dominant cluster, including $v = 0$.

Research question 2 (Singular limiting experiments and optimal constants). Describe the limiting experiments when $a = n^{-1/(4m)}u$ with known zero variance, or $(a, v) = (n^{-1/(4m+4)}u, n^{-1/(2m+2)}t)$ with unknown variance. The exact Hellinger constants identify candidate information coefficients, but do not alone identify the likelihood-ratio process or efficient risk constant. A useful result would include convergence uniform over compact sets of shape parameters and an attainable optimal constant.

Research question 3 (Certified up-information constants and their growth). Develop interval-certified algorithms for $J_r(\text{up})$, beginning with $r = 2, 4, 6$. Monotone prefix values give upper bounds; interior integrals and controlled endpoint estimates can give lower bounds. Prove an effective error bound, then determine the growth of $J_r(\text{up})$ as $r \rightarrow \infty$. Without such control, the fixed-capacity rates cannot be read uniformly in an increasing number of factors.

Research question 4 (The finite-to-infinite crossover). Let the available background prefix length $M = M(h)$ grow while $h \rightarrow 0$, or let the geometric ratio of b_j vary with h . Determine when the finite-boundary term, the critical logarithm, and the full information constant dominate. A solution must track the width of the exact endpoint collar and the size of the information constants; substituting a varying M into a fixed- M asymptotic is not valid.

Research question 5 (Minimal tail hypotheses). All absolute moments suffice in [theorem 2.1](#); sub-Gaussian or exponential tails are unnecessary. For a prescribed order r , determine the weakest finite-moment or tail condition ensuring the same expansion. The truncation proof suggests a tradeoff between available prefix length and available noise moments. Find sharp sufficient conditions and a heavy-tail counterexample at the threshold.

Research question 6 (Beyond-all-powers separation). For the explicit lognormal pair in [theorem 5.2](#), determine the actual first nonzero asymptotic scale of the Hellinger distance. It is smaller than every power of h , but nonzero. Does a logarithmic-square scale analogous to the lognormal tail govern its leading behavior, and how does the background enter the coefficient? This question lies beyond finite-order moment methods by construction.

Research question 7 (Adaptation and honest uncertainty quantification). Construct an estimator that adapts to unknown collision multiplicities, vanishing slots, and positive versus zero variance. Determine whether confidence sets can be simultaneously honest and rate-adaptive on all these strata. Separate an impossibility caused by nearly indistinguishable strata from inefficiency of a particular algorithm.

Research question 8 (Efficient certified moment fitting). Replace the $\binom{n+m}{m}$ grid by an algorithm with explicit dependence on m and optimization tolerance. A successful certificate should guarantee an objective gap $O(n^{-1/2})$, which is sufficient for the proved rates, even when roots coincide or vanish. Hyperbolic-polynomial coefficient methods and interval branch-and-bound are plausible starting points, not implemented solutions in this package.

Research question 9 (Multivariate uniform directions). Replace $a_i V_i$ by $V_i a_i$ with vectors $a_i \in \mathbb{R}^d$, and permit an unknown Gaussian covariance matrix. Higher cumulants become sums of even tensor powers. Determine identifiable parameter classes up to signs and permutations, and then prove a multivariate finite-boundary phase diagram that includes faces of different codimensions. The one-dimensional counting-function inverse inequality does not extend automatically.

Research question 10 (Characterize admissible compact backgrounds). Find a useful intrinsic condition on a compactly supported density g that is equivalent, or close to equivalent, to the all-order Hellinger moment-matching property. Is finite $J_r(g)$ together with an appropriate translation-uniform integrability condition sufficient at each fixed r ? The present uniform-prefix representation is a strong, checkable sufficient condition, but not claimed necessary.

16 Conclusion

For the unsmoothed Fabius–Rvachev law, a moving compact-support boundary does not change the finite-order moment-based Hellinger exponent. The reason is specific and quantitative: sufficiently long uniform prefixes have enough endpoint vanishing to make the relevant score square integrable, their

information converges to the infinite-background information, and common-convolution contraction transfers the sharp asymptotic upper bound. Fatou supplies the matching lower bound.

This closes the identified unsmoothed statistical gap for every fixed finite capacity and extends the lower-bound constructions to the boundary of unknown Gaussian variance. Finite prefixes show why this conclusion was not automatic: too few uniform factors produce a different power, and the exact critical number produces a logarithm. The result is a proved continuation of a specific research program, with its classical algebra and information ingredients distinguished from the analytic and zero-noise conclusions developed here.

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- [7] *Gaussian Confounding and Sharp Recovery of Uniform Convolution Factors: Shifted power sums, minimax estimation, and detection above a Fabius–Rvachev background*, research report prepared for Vladimir Reshetnikov, 29 September 2026, 19 pages. Read from the user’s Library as `article(20260929-161307).pdf`, version 1. Theorem 4.1 supplies the credited shifted-moment inequality; Research Question 1 on page 17 explicitly asks for the unsmoothed up-law extension. Its own repository comparison is pinned to `c130dba623c420551d90db41dae7b3d9cff72dfd`, not the snapshot used here. No publication or independent review is implied.